Reducing costs for NAT gateway

You have to pay for the traffic processed by a NAT gateway (see VPC Pricing at https://aws.amazon.com/vpc/pricing/ for more details). If your EC2 instances in private subnets will have to transfer huge amounts of data to the internet, you have two options to decrease costs:

- Moving your EC2 instances from the private subnet to a public subnet allows them to transfer data to the internet without using the NAT gateway. Use firewalls to strictly restrict incoming traffic from the internet.
- If data is transferred over the internet to reach AWS services (such as Amazon S3 and Amazon DynamoDB), use gateway VPC endpoints. These endpoints allow your EC2 instances to communicate with S3 and DynamoDB directly and at no additional charge. Furthermore, most other services are accessible from private subnets via interface VPC endpoints (offered by AWS PrivateLink) with an hourly and bandwidth fee.

WARNING NAT gateways are finite resources. A NAT gateway processes up to 100 Gbit/s of traffic and up to 10 million packets per second. A NAT gateway is also bound to an availability zone (introduced in chapter 16).

To keep concerns separated, you'll create a subnet for the NAT gateway as follows:

```
SubnetPublicNAT:
  Type: 'AWS::EC2::Subnet'
  Properties:
    AvailabilityZone: !Select [0, !GetAZs '']
    CidrBlock: '10.0.0.0/24'                    ①
    MapPublicIpOnLaunch: true
    VpcId: !Ref VPC
    Tags:
    - Key: Name
      Value: 'Public NAT'
RouteTablePublicNAT:
```